

Embedded systems exp Exp 26

Program

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module
{
    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket")
    {
        socket.register_b_transport(this, &Target::b_transport);
    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay)
    {
        int* data = (int*)trans.get_data_ptr();

        cout << "Target received: " << *data << endl;
```

```

        delay += sc_time(10, SC_NS);
    }
};

// Initiator
struct Initiator : sc_module
{
    tlm_utils::simple_initiator_socket<Initiator> socket;

    SC_CTOR(Initiator) : socket("socket")
    {
        SC_THREAD(thread_process);
    }

    void thread_process()
    {
        tlm_generic_payload trans;

        sc_time delay = SC_ZERO_TIME;

        int data = 555;

        trans.set_data_ptr(
            reinterpret_cast<unsigned char*>(&data)
        );

        trans.set_data_length(sizeof(data));
    }
};

```

```
    cout << "Initiator sending: " << data << endl;

    socket->b_transport(trans, delay);

    wait(delay);
}

};

int sc_main(int argc, char* argv[])
{
    Initiator initiator("initiator");

    Target target("target");

    initiator.socket.bind(target.socket);

    sc_start();

    return 0;
}
```

OUTPUT

```
[2026-08-24 02:25:25 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
```

Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56

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Initiator sending: 555

Target received: 555

Done